

Abhishek Kumar Gupta

Lead Software Development Engineer |
Distributed Systems & AI Architect

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About Me

Lead Software Development Engineer and AI Platform Architect with 10+ years of experience building scalable distributed systems, cloud-native platforms, and enterprise AI solutions across automotive, enterprise, security, and AI domains. Strong expertise in C#, .NET, Python, backend platform engineering, distributed systems, scalable API design, high-throughput event-driven architectures, Azure, and Agentic AI. Proven track record delivering high-availability systems processing 3B+ daily messages, leading cloud modernization initiatives, and building enterprise AI platforms at scale. Strong foundation in scalable system, microservices and data structures & algorithms. Experienced in leading distributed engineering teams, optimizing platform reliability and performance, and collaborating with global stakeholders across the US, EU, and EMEA regions.

Skills

- **Programming Languages & Frameworks:** C#, .NET/ .NET Core, Python, Angular, TypeScript, SQL Server, REST APIs, Microservices, Object-Oriented Programming (OOP), Data Structures & Algorithms, KQL (Kusto Query Language)
- **Cloud & DevOps:** Microsoft Azure, Azure DevOps, CI/CD Pipelines, Azure Kubernetes Service (AKS), Kubernetes, Docker, API Management (APIM), Azure Functions, Azure Web Apps, Azure WebJobs, Azure Event Hub, Azure Service Bus, Azure Storage Account, Azure Cosmos DB, Azure IoT Hub, Azure Data Explorer (Kusto), Infrastructure as Code (IaC), Git, GitHub, Cloud-Native Architecture, Application Insights
- **AI / Generative AI:** Agentic AI, Generative AI (GenAI), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Azure OpenAI, Azure AI Services, Azure AI Search, Microsoft Foundry, Microsoft Agent Framework, LangChain, LangGraph, Prompt Engineering, Vector Databases, Copilot Studio, Microsoft 365 Copilot
- **Architecture & Engineering:** System Design, Distributed Systems, Scalable Backend Systems, Event-Driven Architecture, Scalable Systems Design, High-Availability Systems, Fault-Tolerant Systems, High-Throughput Systems, Secure Software Development, Performance Optimization, Reliability Engineering, Design Patterns, Multithreading, Asynchronous Programming
- **Software Engineering Practices:** Agile/Scrum, Test-Driven Development (TDD), Code Reviews, Secure Coding Practices, Release Management, Production Support
- **Leadership & Collaboration:** Technical Leadership, Engineering Mentorship, Cross-Functional Team Leadership, Stakeholder Management, Global Team Collaboration

Work Experience

Senior Consultant, Microsoft GCID

May 2021- Current

Location: Hyderabad, India

- **Kaseya - AI Digital Worker System** Team Size: 40
 - Architected and developed an **AI-powered Digital Worker platform** using Agentic AI frameworks to automate enterprise ticket triaging and operational workflows.
 - **Reduced ticket triaging effort** by ~87% (12 minutes → 1.5 minutes), significantly improving operational efficiency, SLA adherence, and response times.
 - Designed scalable fault-tolerant **event-driven microservices** and cloud-native containerized services on AKS, enabling resilient, high-throughput distributed workflow orchestration at scale.
 - Implemented **enterprise-grade observability**, telemetry, and monitoring using Azure App Insights, improving platform reliability, diagnostics, and deployment efficiency.
- **Microsoft - AI Code Migration Agent** Team Size: 28
 - Architected and developed an AI-first, **agent-based migration platform** supporting Python, Java, .NET, and Go, reducing AWS Lambda to Azure Functions migration **effort by ~75%**.
 - Designed a unified code and unit test migration framework **enabling >90% unit test reuse**, significantly reducing validation effort and post-migration defects.
 - Built scalable high-throughput **semantic code intelligence systems** using Microsoft Agent Framework, RAG, vector search, and intelligent code chunking, improving **large-repository analysis accuracy** by ~80% and reducing migration rework by ~75%.
 - Developed **cloud-native AI services** on AKS with Azure DevOps CI/CD pipelines and **enterprise-grade observability**, enabling scalable, secure, and production-ready deployments.
- **Roche – MOVE Compass AI Assistant** Team Size: 100
 - Architected and developed an **AI-powered enterprise Migration Assistant** leveraging **Copilot Studio**, **MCP**, **Azure**, and **Graph**

Connectors, supporting **120K+ enterprise users** across multiple business functions.

- Enabled **scalable contextual AI experiences** using **500M+** Google Drive file contexts and ServiceNow enterprise knowledge integration, significantly improving response relevance and knowledge accessibility.
- Designed **AI-guided onboarding** and **support workflows** enabling seamless pre-, during-, and post-migration user assistance.
- Published and integrated the AI Assistant across **Web, Microsoft 365 Copilot, Microsoft Teams, and SharePoint**, driving seamless multi-channel adoption and enterprise accessibility.

- **Renault-Nissan Alliance – AvNext Connected Car Platform**

Team Size: 200

- Led development and production releases for a large-scale connected car platform across multiple environments, delivering **~20 consecutive releases** with **zero Critical/High severity incidents**.
- Architected and scaled high-throughput, cloud-native event-driven platforms across **8+ enterprise products**, enabling resilient and fault-tolerant distributed processing at scale.
- **Reduced** per-vehicle **platform operational cost** by ~63% (€0.54 → €0.20) through scalable architecture modernization, performance optimization, and efficient workload processing.
- Led architecture and **delivery** for a **24-engineer team** across **60+ Agile sprints**, driving platform reliability, observability, security, and near-zero Sev-1/Sev-2 defects.

Research & Development Engineer, ABB

September 2020 – April 2021

Location: Bengaluru, India

- **EL-BackOffice**

Team Size: 12

- Contributed to the **design** and **development** of **scalable cloud-native** microservices applications using C#, .NET Core, Angular, SQL Server, Cosmos DB, and Azure, delivering resilient enterprise solutions with high reliability.
- Developed **reusable** Angular and .NET Core frameworks, **improving code quality**, delivery efficiency, and engineering standards through CI/CD automation, unit testing, and peer code reviews.

Deputy Manager, Deloitte Tax Technology

October 2018 – September 2020

Location: Bengaluru, India

- **TP Digital DoX**

Team Size: 40

- Contributed to the architecture and development of **scalable cloud-native** tax technology platforms using C#, .NET Core microservices, Angular, SQL Server, MongoDB, Redis, and Azure.
- **Built reusable** Angular and .NET Core frameworks, improving **engineering efficiency**, code quality, and deployment reliability through CI/CD automation, unit testing, and code reviews.

Software Engineer, 42Gears Mobility Systems

August 2016 – October 2018

Location: Bengaluru, India

- **SureMDM**

Team Size: 20

- Contributed to the **development** and **enhancement** of SureMDM, a scalable enterprise mobility and endpoint management platform using C#, .NET, Angular, SQL Server, REST APIs, and Azure.
- Designed and implemented secure, scalable features across **Admin, SSO, OAuth, Chrome OS**, and Security modules, improving enterprise device administration and access management capabilities.

Education

- Master of Technology, Computing Systems & Infrastructure - Birla Institute of Technology and Science | CGPA: 8.39/10
- Bachelor of Engineering, Information Science & Engineering – PESIT (PES University) | CGPA: 8.78/10.0 (First Class with Distinction all 8 semesters)

Certifications

- Azure Solutions Architect Expert (AZ-305)
- DevOps Engineer Expert (AZ-400)
- Azure AI Engineer Associate (AI-102)
- Azure Cosmos DB Developer Specialty
- GitHub Copilot (GH-300)
- GitHub Advanced Security (GH-500)

Awards

- Microsoft Distinguished Gold Club Award 2025 – Outstanding contributions to RNA project and organizational impact.
- Microsoft Global Hackathon 2025 Winner – Built AI-driven enterprise operational excellence solution.
- Microsoft Rise Award 2025 – High-impact engineering contributions to RNA project.
- Microsoft Best Lead Award 2024 – Recognized for exceptional technical leadership, team mentorship, and delivery execution.
- Deloitte Commitment to Talent Award 2019 – Recognized for driving effective teamwork and seamless production releases.
- Deloitte Commitment to Talent Award 2020 – Awarded for end-to-end application development and customer onboarding.